

Dynamic Guideline: WHO-Based Injuries - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope of this report

This report addresses **injuries and violence** as a broad WHO public-health and clinical-management domain, including:

- Road traffic injuries
- Drowning
- Falls
- Burns
- Poisoning
- Violence-related injuries, including homicide, self-harm/suicide, and war/conflict-related injury
- Trauma care, rehabilitation, surveillance, and policy management

WHO frames injuries and violence as a major preventable global health burden. WHO reports that injuries and violence account for about **4.4 million deaths annually**, including **3.16 million deaths from unintentional injuries** and **1.25 million deaths from violence-related injuries** each year [WHO injuries and violence fact sheet](#). WHO also reports that injuries and violence caused **4.4 million deaths in 2019**, representing **8% of all deaths** [WHO Preventing injuries and violence overview](#).

Verification note on the requested “Meaningful youth engagement” title

The WHO publication matching the requested title is “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**”, published/launched on **11 December 2025** [WHO publication page](#), [WHO webinar page](#). It is **not an injuries guideline**; it is a WHO evidence brief using the **WHO Abortion care guideline (2022)** as the case study [WHO Abortion care guideline](#). Therefore, it is considered in this report only as **guideline-development methodology evidence** for equity, inclusivity, and meaningful youth engagement, and is **not counted** among the five WHO injury-related guidelines below.

Exactly 5 WHO injury-related guidelines / technical guidance documents synthesized

No.	WHO guideline / guidance document	Publication date available from supplied evidence	Target population / users	Scope and objectives
1	Developing policies to prevent injuries and violence: guidelines for policy-makers and planners	10 November 2006	Policy-makers, planners, ministries of health, multisectoral injury-prevention stakeholders	Cross-cutting policy guideline for developing national and subnational injury-and-violence prevention policies WHO publication page .
2	Guidelines for essential trauma care	WHO injury-control guidance; exact publication date not specified in supplied summary	Health systems, trauma clinicians, emergency-care planners	Minimum essential trauma-care resources, organization, triage, resuscitation, referral, and hospital-based care priorities WHO IRIS PDF .
3	Injury surveillance guidelines	WHO surveillance guidance; exact publication date not	Public-health surveillance teams, health ministries,	Standardized collection and classification of injury-event,

No.	WHO guideline / guidance document	Publication date available from supplied evidence	Target population / users	Scope and objectives
		specified in supplied summary	emergency departments	mechanism, intent, setting, and outcome data WHO IRIS PDF .
4	WHO guideline on the prevention of drowning through provision of day-care and basic swimming and water safety skills	Date not shown in supplied snippet	Governments, child-health programmes, education and water-safety sectors	Prevention of drowning through safe day-care, basic swimming, water-safety skills, and safe rescue approaches; listed among WHO drowning-prevention guidance WHO overview PDF mirror .
5	Step safely: strategies for preventing and managing falls across the life-course	2021	Policy-makers, clinicians, older-adult services, occupational-health and urban-design stakeholders	Life-course falls prevention and management, including risk-factor reduction, awareness, environmental safety, clinical prevention, and management WHO overview PDF mirror .

Primary anchor guideline for this dynamic report: Developing policies to prevent injuries and violence: guidelines for policy-makers and planners, published by WHO on **10 November 2006**, listed by WHO as a **Guideline**, 74 pages, ISBN **9241593504** [WHO publication page](#). Newer WHO technical packages and fact sheets update the implementation context but do not replace this cross-cutting injury-and-violence policy guideline.

1.2 Key Recommendations

GRADE note: The 2006 WHO policy guideline and several older WHO technical documents do not provide formal GRADE evidence profiles in the supplied evidence. The GRADE ratings below are therefore **analyst-applied** using standard GRADE principles: risk of bias, consistency, directness, precision, publication bias, effect size, plausibility, feasibility, equity, resource use, and balance of benefits and harms.

Recommendation	GRADE Evidence Quality	Strength
Establish a national or subnational injury-and-violence prevention policy with clear leadership, measurable objectives, funding, accountability, and multisectoral governance.	Moderate	Strong
Use injury surveillance systems to define burden, identify high-risk groups, monitor trends, evaluate policy effects, and guide resource allocation.	Moderate	Strong
Prioritize interventions for the largest contributors to injury death and disability: road traffic crashes, falls, drowning, burns, poisoning, suicide, homicide, and conflict-related injuries.	Moderate	Strong
Address social inequities in injury risk, including higher risk among children, young people, males, low-income populations, and marginalized communities.	Moderate	Strong
Implement evidence-informed road-safety policies , including speed management, safer infrastructure, protection of pedestrians/cyclists/motorcyclists, seat belts, helmets, drink-driving countermeasures, and vehicle-safety standards.	Moderate	Strong
Implement drowning-prevention strategies, including barriers to water, safe child supervision/day-care, basic swimming and water-safety skills, safe rescue, boating regulation, and flood-risk management.	Low–Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
Implement burn-prevention strategies addressing unsafe cooking, heating, electrical hazards, flammable substances, first-aid training, burn-care capacity, and burn registries.	Low–Moderate	Strong
Implement falls prevention across the life-course, especially for older adults, including medication review, environmental modification, balance/strength interventions, vitamin D deficiency management where relevant, and occupational/urban-design safety.	Moderate	Strong
Integrate essential trauma care into injury policy, including prehospital response, validated triage, resuscitation, surgical stabilization, referral pathways, rehabilitation, and special-population care.	Moderate	Strong
Include young people and affected communities in injury-policy development through meaningful, non-tokenistic engagement, especially for adolescent and youth injury-prevention programmes.	Low	Conditional

1.3 Guideline Development Methodology

Evidence review process

WHO defines a guideline as any WHO information product containing recommendations for clinical practice or public-health policy [WHO Guidelines](#). WHO recommendations are intended to help users decide **whether, when, and how** to undertake clinical interventions, diagnostic tests, or public-health measures to achieve the best health outcomes [WHO Guidelines](#).

For the 2006 injury-and-violence policy guideline, the supplied evidence indicates that the document was designed for **policy-makers and planners**, was drafted to avoid being overly prescriptive, and used country experiences and case studies to illustrate policy-development processes [WHO injury policy PDF](#).

Consensus method

The guideline approach is consistent with WHO public-health policy guidance: problem definition, stakeholder engagement, evidence-informed intervention selection, multisectoral planning, implementation, monitoring, and evaluation. WHO’s broader guideline governance states that the WHO Guidelines Review Committee ensures WHO guidelines meet methodological quality standards through transparent, evidence-based decision-making [WHO Guidelines](#).

Conflict of interest management

The supplied evidence does not provide a specific conflict-of-interest table for the 2006 injury-and-violence policy guideline. Current WHO guideline governance requires rigorous quality assurance and transparent evidence-based processes [WHO Guidelines](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence considered after the 2006 WHO anchor guideline included:

- WHO injury, violence, trauma, road-safety, drowning, falls, and burns documents
- WHO fact sheets and status reports
- CDC road-safety and traumatic brain injury resources

- NICE rehabilitation and falls-prevention evidence
- Recent systematic reviews and meta-analyses through 2025
- Trauma-management guideline updates from major clinical organizations
- WHO youth-engagement guideline-development evidence brief for methodology relevance only

Databases and sources represented in supplied evidence

- WHO publications and WHO fact sheets
- WHO IRIS
- CDC
- NICE / NCBI Bookshelf
- PubMed
- Journal of Road Safety
- American College of Surgeons
- EAST trauma practice-management guideline catalog
- Peer-reviewed and indexed systematic reviews where available

Inclusion criteria

Included evidence met at least one of the following:

1. WHO source related to injury prevention, trauma care, surveillance, or violence prevention.
2. Post-2006 injury-prevention or injury-management evidence.
3. Systematic review, meta-analysis, guideline, surveillance report, or authoritative public-health document.
4. Evidence relevant to diagnosis, treatment, management, prevention, rehabilitation, surveillance, or policy.

Exclusion criteria

Excluded or downgraded evidence if:

- Not injury-related.
- Not from WHO or another authoritative public-health/medical source.
- No extractable findings were available.
- Only a title or metadata was available without outcome details.

2.2 Key New Studies and Evidence

Study / source	Design / type	Key findings	GRADE Quality
WHO Preventing injuries and violence: an overview	WHO public-health overview	Injuries and violence caused 4.4 million deaths in 2019 , representing 8% of all deaths ; key causes include road traffic crashes, drowning, falls, burns, poisoning, and violence WHO overview .	Moderate
WHO Injuries and violence fact sheet	WHO burden and prevention synthesis	Annual injury-related deaths include 3.16 million unintentional injury deaths and 1.25 million violence-related deaths ; three of the top five causes of death among people aged 5–29 years are injury-related WHO fact sheet .	Moderate
WHO Global status report on road safety 2023	Global status report	Annual road traffic deaths decreased slightly to 1.19 million , but road traffic injuries remain the leading killer of children and young people aged 5–29 years;	Moderate

Study / source	Design / type	Key findings	GRADE Quality
		pedestrians, cyclists, and motorcyclists account for more than half of deaths WHO road safety report .	
2024 systematic review: Effectiveness of interventions for preventing road traffic injuries	Systematic review	Included studies through December 2023; traditional meta-analysis not feasible because of heterogeneity; structured synthesis supported multiple road-safety intervention categories PMC .	Low– Moderate
NCBI Bookshelf road-traffic injury synthesis	Evidence synthesis	Speed cameras, vehicle safety technologies, and electronic stability control were associated with reductions in fatalities or injuries; hazard-elimination approaches were more effective than relying only on behavior change NCBI Bookshelf .	Moderate
CDC pedestrian safety surveillance	Surveillance analysis	U.S. pedestrian death rates increased 50% from 2013 to 2022, while 27 other high-income countries generally decreased, median 24.7% decrease CDC .	Moderate
CDC teen-driver risk factors	Public-health guidance / surveillance	Motor vehicle crashes are a leading cause of death for U.S. teens; risk factors include inexperience, night/weekend driving, seat-belt nonuse, distraction, speeding, alcohol, and drugs CDC teen drivers .	Moderate
WHO burns fact sheet	WHO fact sheet	Burns are preventable; high-income countries reduced burn deaths through prevention and care improvements, but implementation gaps remain in LMICs WHO burns fact sheet .	Low– Moderate
2025 systematic review/meta-analysis: clinical decision support for falls prevention in older adults	Systematic review and meta-analysis	Searched through January 2025; evaluated CDS interventions for fall prevention in adults ≥65 years; certainty rated using GRADE PubMed .	Moderate
NICE falls evidence review methods, April 2025	Guideline evidence-review methods	Uses GRADE-adapted approach considering risk of bias, indirectness, inconsistency, imprecision, and publication bias NCBI Bookshelf PDF .	Moderate
2014–2025 road-traffic injury severity prediction review	Systematic review and meta-analysis	Machine-learning/deep-learning prediction of crash injury severity; certainty for primary outcome rated moderate using GRADE criteria adapted for prognostic studies Journal of Road Safety .	Moderate
WHO improving care of the injured	WHO activity / technical guidance	Emphasizes validated acuity-based triage, resuscitation, surgical stabilization, referral systems, and special-population needs WHO improving care .	Moderate
ACS 2024 revised TBI best-practice guideline	Clinical best-practice guideline	Provides updated TBI care guidance; CT indications, ICP threshold around 22 mm Hg , CPP target ≥60 mm Hg , and PbtO ₂ target ≥15 mm Hg if monitored ACS TBI guideline PDF .	Moderate
CDC adult mild TBI management resources	Clinical guidance tools	Provides adult mild TBI diagnosis/management checklist, discharge instructions, recovery tips, and return-to-work instructions CDC mTBI .	Moderate

Study / source	Design / type	Key findings	GRADE Quality
NICE rehabilitation after traumatic injury	Clinical guideline	Recommends early multidisciplinary rehabilitation planning, referral to specialist rehabilitation when needed, outcome monitoring, discharge planning, and community reintegration NICE NG211 .	Moderate
EAST 2024–2025 trauma practice-management guideline catalog	Systematic review/meta-analysis-based trauma guidelines	Includes 2024 whole-blood resuscitation and handoffs/transitions guidelines, and 2025 trauma-in-pregnancy guideline EAST .	Low–Moderate
WHO evidence brief on meaningful youth engagement	Guideline-development evidence brief	Published 11 December 2025 ; shows how young people can contribute lived experience and equity perspectives to WHO guideline development, but case study is abortion care, not injuries WHO publication .	Low

2.3 GRADE Evidence-to-Recommendation Synthesis

Clinical / public-health question	Evidence summary	Benefits	Harms / burdens	Equity considerations	GRADE Quality	Recommendation Strength
Should governments maintain national injury-and-violence prevention policies?	WHO identifies injuries and violence as responsible for millions of deaths annually and preventable through evidence-informed policy WHO fact sheet .	Large mortality and disability reduction potential.	Requires sustained funding and intersectoral governance.	High relevance for youth, males, low-income groups, and LMICs.	Moderate	Strong
Should injury surveillance be standardized?	WHO surveillance guidance supports classification of mechanism, intent, place, activity, and outcome WHO IRIS PDF .	Enables targeting and evaluation.	Data-system burden.	Improves visibility of marginalized and high-risk groups.	Moderate	Strong
Should road-safety policy prioritize speed control and safe-system design?	WHO 2023 road-safety report shows persistent 1.19 million deaths; systematic review and NCBI synthesis support	High potential reduction in deaths and serious injuries.	Political and enforcement challenges.	Protects pedestrians, cyclists, motorcyclists, and LMIC populations.	Moderate	Strong

Clinical / public-health question	Evidence summary	Benefits	Harms / burdens	Equity considerations	GRADE Quality	Recommendation Strength
	infrastructure, speed, and vehicle-safety interventions WHO road safety report , NCBI Bookshelf .					
Should drowning prevention include safe child supervision/day-care and swimming/water-safety skills?	WHO lists drowning as a major injury cause and identifies drowning-prevention guidance and implementation resources WHO fact sheet , WHO overview PDF mirror .	Prevents child drowning deaths.	Requires trained staff, safe facilities, and local adaptation.	High relevance for children in flood-prone and water-exposed communities.	Low–Moderate	Strong
Should burn prevention combine policy, education, safer environments, first aid, and registry-based monitoring?	WHO states burns are preventable and recommends multisectoral burn-prevention planning, community first-aid training, burn-care strengthening, and global burn registry use WHO burns fact sheet .	Reduces mortality, disability, and scarring.	Implementation costs.	Important for women/girls in LMICs exposed to unsafe cooking and energy poverty.	Low–Moderate	Strong
Should older-adult falls prevention include risk assessment and decision support?	2025 review evaluated clinical decision support for fall prevention using GRADE; NICE 2025 methods use GRADE-adapted evidence review PubMed , NCBI Bookshelf PDF .	Reduces fall risk and potentially fall injuries.	Alert fatigue and workflow burden possible.	High relevance for ageing populations and long-term care.	Moderate	Strong

Clinical / public-health question	Evidence summary	Benefits	Harms / burdens	Equity considerations	GRADE Quality	Recommendation Strength
Should trauma systems ensure validated triage, resuscitation, surgical stabilization, referral, and rehabilitation?	WHO improving-care guidance and NICE trauma rehabilitation recommend triage, resuscitation, referral, multidisciplinary rehabilitation, and discharge planning WHO improving care , NICE NG211 .	Reduces preventable death and disability.	Requires trained workforce and system capacity.	Benefits rural, low-resource, conflict, and disaster settings if referral pathways are equitable.	Moderate	Strong
Should TBI management use structured imaging, monitoring, discharge, and rehabilitation protocols?	ACS 2024 TBI guideline, CDC mTBI tools, and NICE rehabilitation guidance support structured care ACS TBI guideline PDF , CDC mTBI , NICE NG211 .	Improves diagnosis, monitoring, recovery, and return to function.	Over-imaging risk if criteria poorly applied.	Supports children, older adults, pregnant people, and those with limited access to follow-up.	Moderate	Strong
Should youth be meaningfully engaged in injury-prevention policy?	WHO 2025 evidence brief supports youth engagement in guideline development, though case study is abortion care, not injury WHO publication .	Improves relevance, acceptability, equity, and uptake.	Requires compensation, safeguarding, and anti-tokenism processes.	Particularly relevant because injuries are leading causes of youth mortality.	Low	Conditional

2.4 Updated Recommendations Based on New Evidence

Original WHO policy recommendation area	New evidence impact	Updated recommendation
Develop national injury-and-violence prevention policies.	WHO burden data confirm injuries and violence remain responsible for about 4.4 million deaths annually WHO fact sheet .	Maintain or establish funded national injury-prevention strategies with measurable targets, legal authority, multisectoral governance, and annual reporting.

Original WHO policy recommendation area	New evidence impact	Updated recommendation
Use surveillance to guide policy.	WHO surveillance guidance and city-level monitoring emphasize standardized indicators and evidence-based decision-making WHO IRIS PDF , WHO city-level monitoring .	Implement interoperable injury surveillance across emergency departments, trauma registries, police, transport, occupational, and mortality systems.
Prevent road traffic injuries.	WHO 2023 report shows road traffic deaths remain high at 1.19 million annually and concentrated among vulnerable road users WHO road safety report .	Adopt a Safe System approach: speed management, safer roads, safer vehicles, safer users, post-crash care, and protection for pedestrians, cyclists, and motorcyclists.
Prevent child and adolescent injury.	WHO reports three of the top five causes of death among ages 5–29 are injury-related WHO fact sheet .	Prioritize youth injury prevention, including road safety, drowning prevention, violence prevention, suicide prevention, school/community safety, and meaningful youth engagement.
Prevent drowning.	WHO identifies drowning as a major child mortality cause and lists drowning-prevention guidance WHO fact sheet .	Scale safe child supervision/day-care, barriers to water, basic swimming and water-safety skills, safe rescue/resuscitation, personal flotation devices, boating regulation, and flood-risk planning.
Prevent falls.	2025 GRADE-linked falls evidence supports structured falls prevention and clinical decision support in older adults PubMed .	Integrate falls risk assessment, medication review, strength/balance programmes, home/environmental modification, vision/footwear review, and clinical decision support into older-adult care.
Prevent burns.	WHO states burns are preventable and calls for multisectoral burn prevention, first-aid education, burn-care strengthening, and registry use WHO burns fact sheet .	Implement national burn-prevention plans targeting cooking, heating, electricity, flammable liquids, occupational hazards, first aid, acute burn care, rehabilitation, and registry-based monitoring.
Strengthen trauma care.	WHO, ACS, CDC, NICE, and EAST evidence supports triage, resuscitation, TBI protocols, transitions of care, and rehabilitation WHO improving care , ACS TBI guideline PDF , NICE NG211 .	Establish integrated trauma systems: emergency access, prehospital care, validated triage, definitive surgical care, TBI protocols, referral networks, rehabilitation, and quality improvement.
Address inequities.	WHO states injury risk is socially patterned and higher among young, male, and low-socioeconomic-status groups WHO overview .	Require equity impact assessment for all injury policies and allocate resources to high-risk populations, LMIC settings, rural areas, conflict-affected populations, and marginalized communities.
Include affected communities.	WHO 2025 youth-engagement evidence brief supports lived-experience input in guideline development WHO publication .	Use meaningful community and youth engagement with power-sharing, safeguarding, compensation, and feedback loops in injury-policy and programme design.

2.5 Practical Clinical and Public-Health Management Pathway

A. Prevention and policy management

- 1. Establish a national injury-prevention strategy.
- 2. Build surveillance and trauma registries.
- 3. Prioritize high-burden mechanisms: road traffic, falls, drowning, burns, poisoning, violence.
- 4. Use legislation, environmental design, engineering, enforcement, education, and emergency-care strengthening.
- 5. Monitor equity and outcomes annually.

B. Diagnosis and acute management of injured patients

- 1. Use validated triage tools and rapid recognition of life-threatening injury, as emphasized by WHO care-of-the-injured guidance [WHO improving care](#).
- 2. Provide immediate resuscitation and hemorrhage control.
- 3. Identify traumatic brain injury, spinal injury, thoracoabdominal injury, burns, poisoning, drowning-related hypoxia, and violence-related injury.
- 4. For suspected TBI, use structured CT criteria and monitoring protocols; ACS 2024 guidance identifies CT indications and monitoring targets such as ICP around **22 mm Hg** and CPP **≥60 mm Hg** in severe TBI contexts [ACS TBI guideline PDF](#).
- 5. Ensure referral to definitive surgical, burn, neurosurgical, toxicology, mental-health, safeguarding, or rehabilitation services as appropriate.

C. Rehabilitation and long-term management

- 1. Start rehabilitation planning early.
- 2. Use multidisciplinary teams.
- 3. Refer promptly for specialist rehabilitation when indicated.
- 4. Monitor patient-reported and clinician-reported outcomes.
- 5. Plan safe discharge and community reintegration, consistent with NICE traumatic-injury rehabilitation guidance [NICE NG211](#).

2.6 Evidence Gaps and Future Research Needs

Gap	Why it matters	Priority research need
Limited recent GRADE-rated evidence for drowning, burns, poisoning, and violence prevention.	These are major causes of death and disability, but the supplied recent GRADE evidence is strongest for falls and road traffic.	Updated systematic reviews and implementation trials using GRADE.
Heterogeneity in road-safety intervention studies.	The 2024 road-traffic review could not conduct traditional meta-analysis because of heterogeneity PMC .	Standardized outcomes, exposure measures, and equity subgroup analyses.
Limited LMIC trauma-system evidence.	Injury deaths and disability are concentrated in lower-resource settings.	Pragmatic trauma-system trials, registry-based evaluations, and cost-effectiveness studies.
Insufficient evidence on community and youth engagement in injury guideline development.	Youth are heavily affected by injuries, but WHO’s 2025 youth-engagement case study is not injury-specific WHO publication .	Injury-specific youth-engagement guideline-development case studies.
Underdeveloped poisoning-prevention evidence in supplied sources.	Poisoning is included in WHO injury burden but detailed recent evidence was limited.	Poisoning surveillance, packaging regulation, toxicology access, and child-poisoning prevention trials.

Gap	Why it matters	Priority research need
Need for better rehabilitation outcome data.	Survival without functional recovery leaves substantial disability burden.	Long-term functional, mental-health, return-to-work, and quality-of-life outcome studies.

2.7 Final WHO-Based Dynamic Recommendations as of 29 April 2026

Recommendation	GRADE Evidence Quality	Strength
Treat injuries and violence as a preventable, high-burden public-health priority requiring national policy, funding, legislation, and multisectoral governance.	Moderate	Strong
Implement standardized injury surveillance and trauma registries to guide prevention, clinical care, and evaluation.	Moderate	Strong
Prioritize road-safety Safe System interventions, especially speed management and protection of pedestrians, cyclists, motorcyclists, children, and young people.	Moderate	Strong
Scale drowning prevention through safe supervision/day-care, barriers to water, swimming and water-safety skills, safe rescue, boating regulation, and flood-risk management.	Low–Moderate	Strong
Scale falls prevention, especially for older adults, using risk assessment, medication review, environmental modification, exercise/balance programmes, and clinical decision support.	Moderate	Strong
Implement burn-prevention plans addressing household, occupational, electrical, cooking, heating, and first-aid risks, with burn registries and strengthened burn care.	Low–Moderate	Strong
Strengthen essential trauma care systems, including prehospital response, triage, resuscitation, surgery, referral, rehabilitation, and quality improvement.	Moderate	Strong
Use structured TBI protocols for diagnosis, imaging, monitoring, discharge, and rehabilitation.	Moderate	Strong
Embed equity, social determinants, and meaningful community/youth engagement into injury-prevention policy and guideline development.	Low–Moderate	Conditional
Update WHO injury-and-violence guidance using formal GRADE evidence profiles, especially for drowning, burns, poisoning, violence prevention, and LMIC trauma systems.	Low	Conditional